IN4MATX 285:
Interactive Technology Studio

Programming: Variables and Conditionals in JavaScript

Today’s goals

By the end of today, you should be able to...

- Conduct basic debugging activities in JavaScript
- Create code in JavaScript which follows programming syntax
- Create variables which hold values and use them to perform actions
- Execute code conditional on values

Disclaimer

- This class is introducing a <u>lot</u> of programming concepts, and quickly
- We don’t expect you to pick up everything immediately
- Assignments will help practice these concepts
  - Online resources and office hours aim to help supplement

Language Roles

HTML

Specify how content is rendered

CSS

Visually style content

JS

Dynamically manipulate content

Language Roles

HTML

Markup
language

CSS

Styling
language

JS

Programming
language

Why JavaScript?

- Make pages dynamic
- Make pages personalized
- Make pages interact
  with other sources,
  like databases and Application
  Programming Interfaces (APIs)

So, how do we use JavaScript?

Loading JavaScript

```
<html>
 <head>
  <script src="test.js"></script>
 </head>
</html>
```

- Separate file, just like CSS

Hello, World!

- Typically the first program you produce in a programming language
- Simply prints the phrase “Hello, World!”

Hello, World!

```
console.log("Hello, world!");
```

Calling a function, more on that later

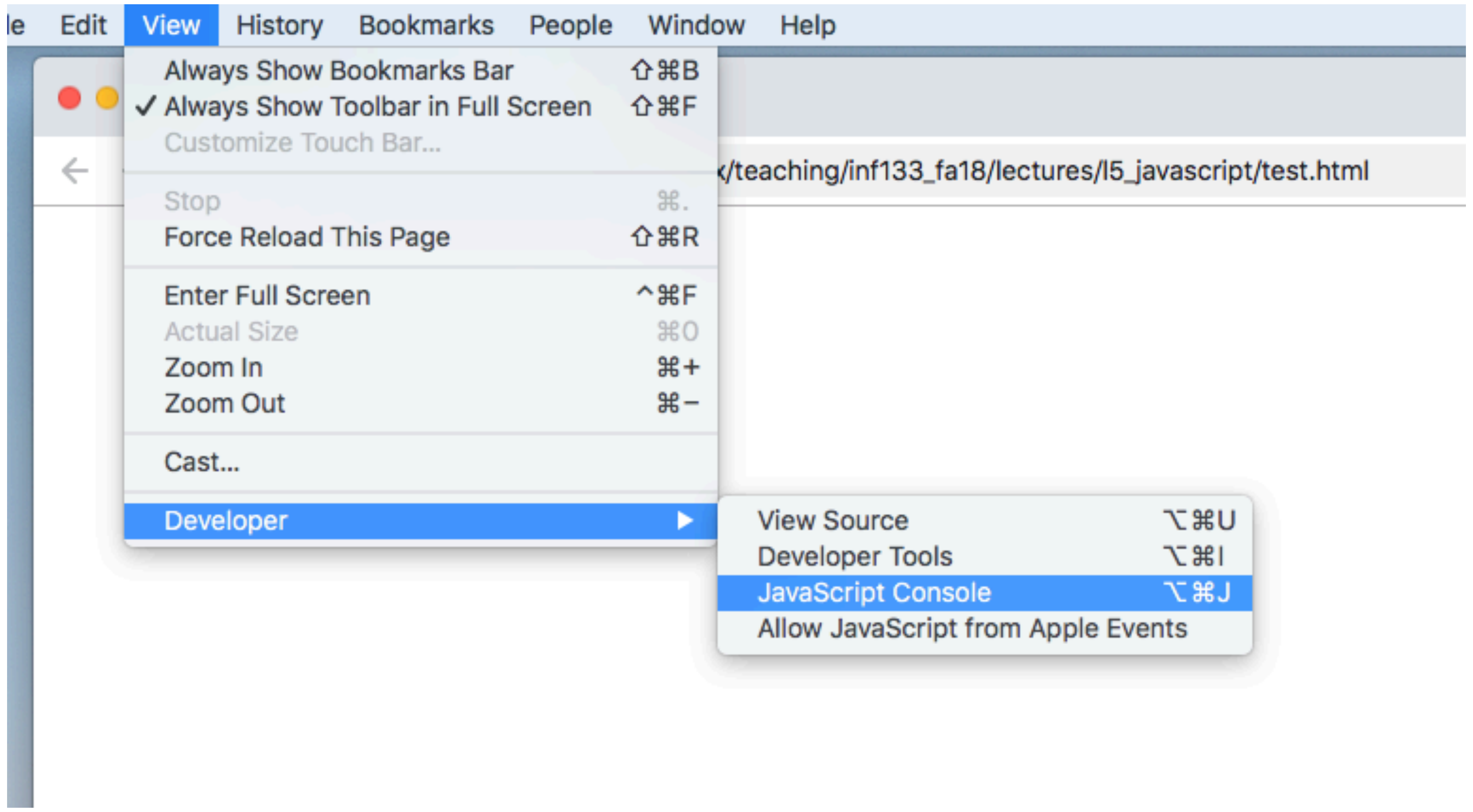
The text to print out

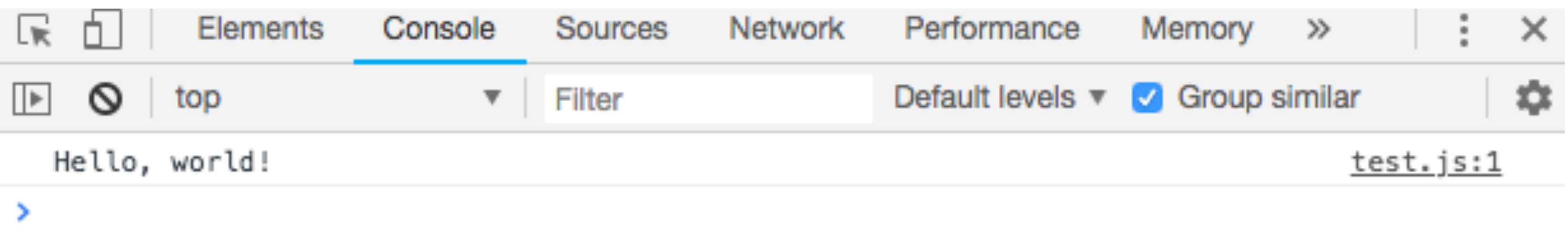
Optional, but often included

Hello, World!

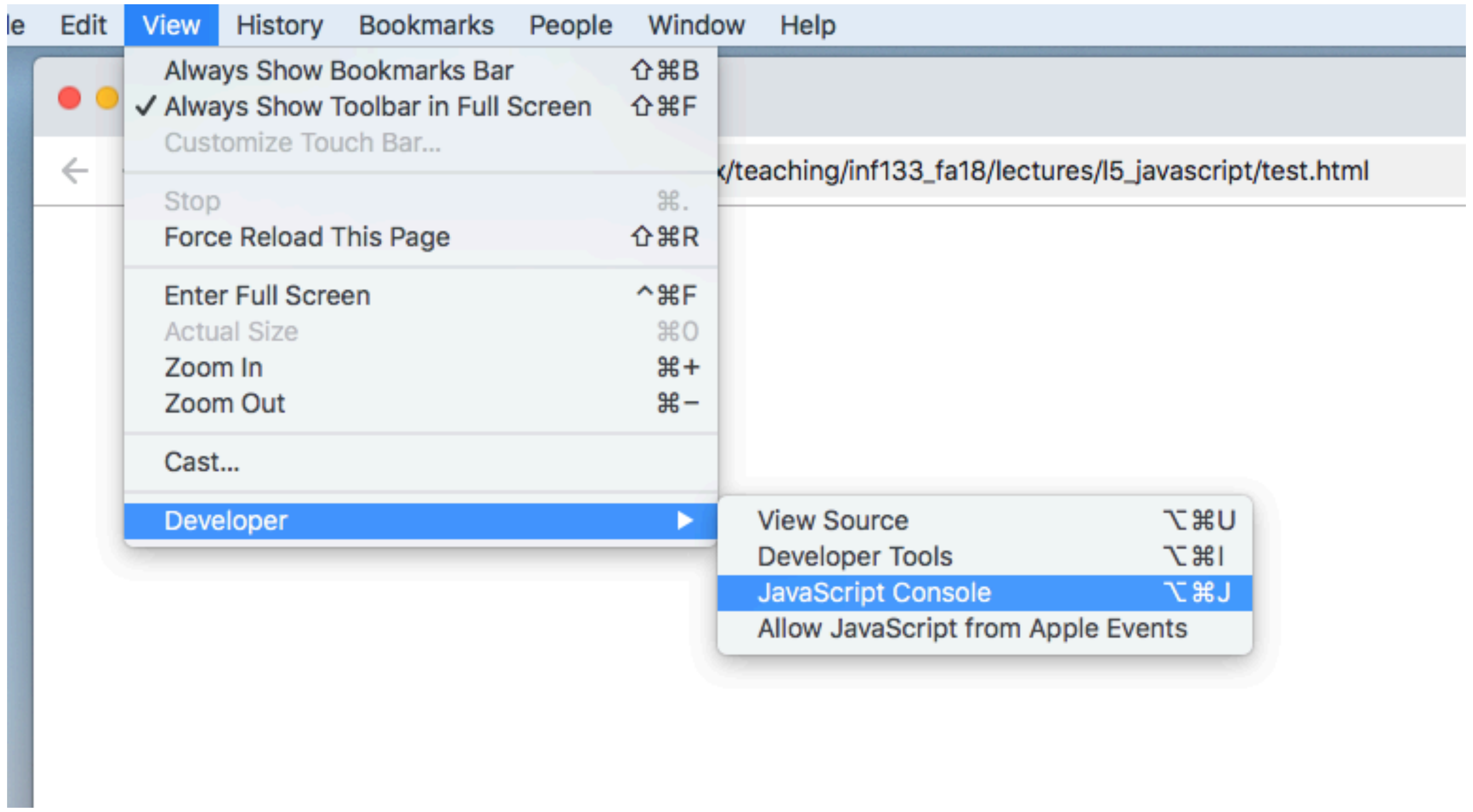
```
console.log("Hello, world!");
```

- Won’t be visible in the browser
- Shows in the JavaScript Console





Hello, World!


le Edit View History Bookmarks People Window Help
Always Show Bookmarks Bar ⇧⌘B
✓ Always Show Toolbar in Full Screen ⇧⌘F
Customize Touch Bar...
Stop ⌘.
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Enter Full Screen ^⌘F
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Zoom In ⌘+
Zoom Out ⌘−
Cast...
Developer
View Source ⌥⌘U
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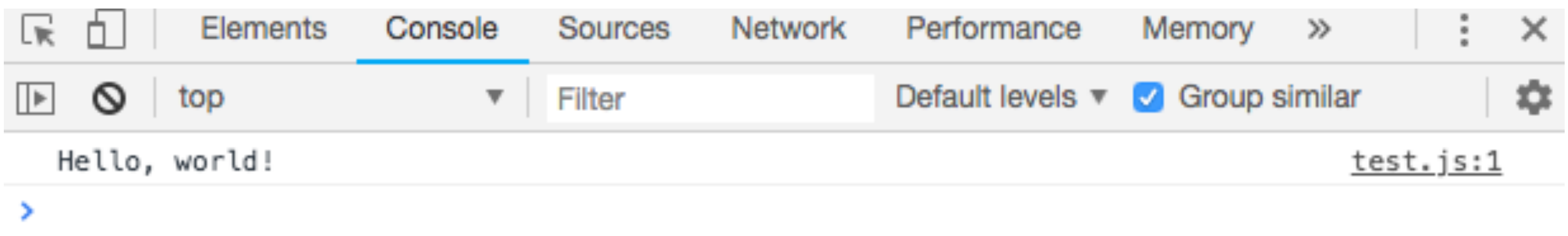
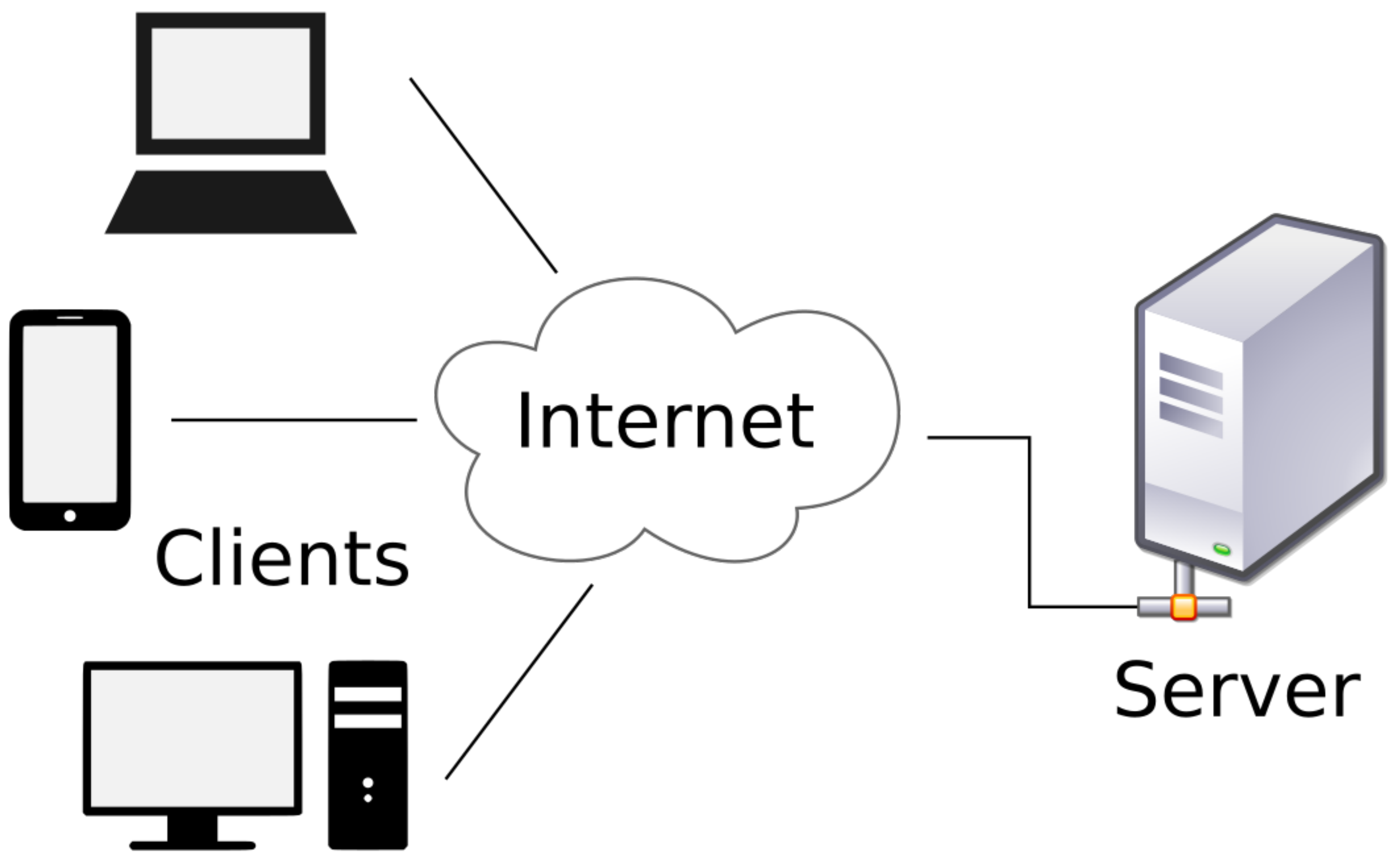
/teaching/inf133_fa18/lectures/l5_javascript/test.html


Elements Console Sources Network Performance Memory
top
Filter
Default levels
Group similar
Hello, world!
test.js:1

Hello, World!

- Why doesn't the JavaScript show up on your page?
- One reason: the HTML file is coming from a **server**, while JavaScript is running on the **client**



Comments

```
//Prints "Hello, world!"
console.log("Hello, world!");
```

- `//` designates that a line should not be executed
- Useful for writing plain-text description of what your code does

Now, how do we make JavaScript useful?

Making JavaScript Useful

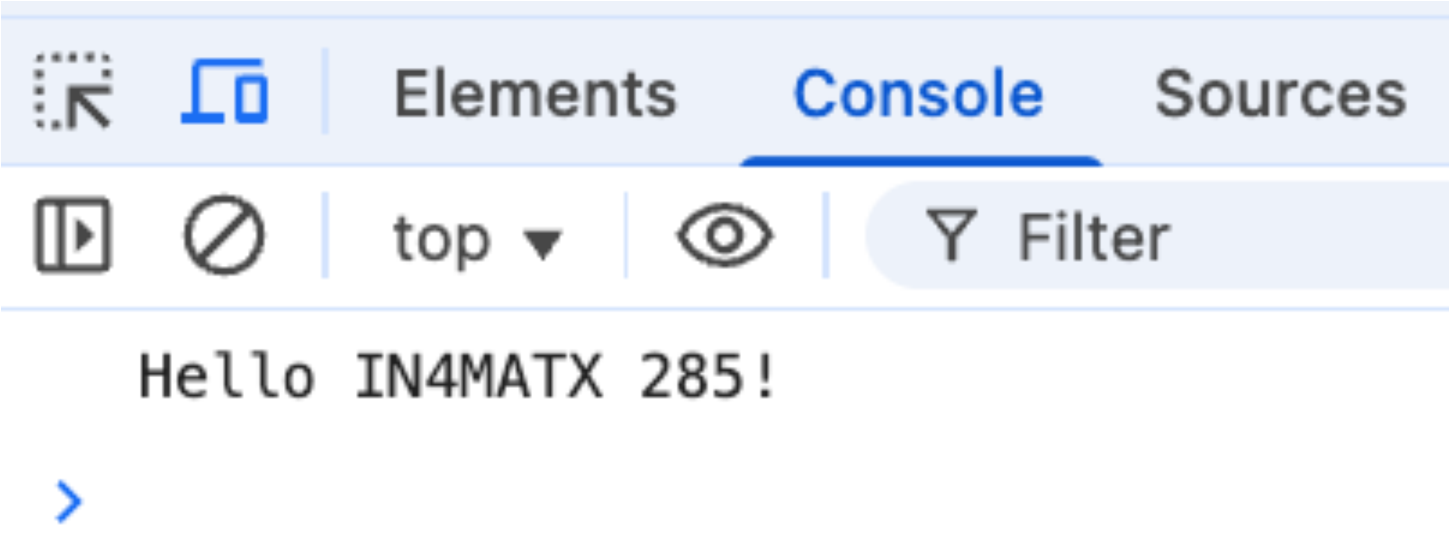
- Variables
- Conditionals
- Loops (next time)
- Functions (next time)

Variables

Variables

- Store values
- Specified with `let`, can be named whatever you want (for the most part)
- Use equals (=) to *assign* a value to a variable
- Can be *referenced* later

```
let variable = 285;
console.log('Hello IN4MATX ' + variable + '!');
```

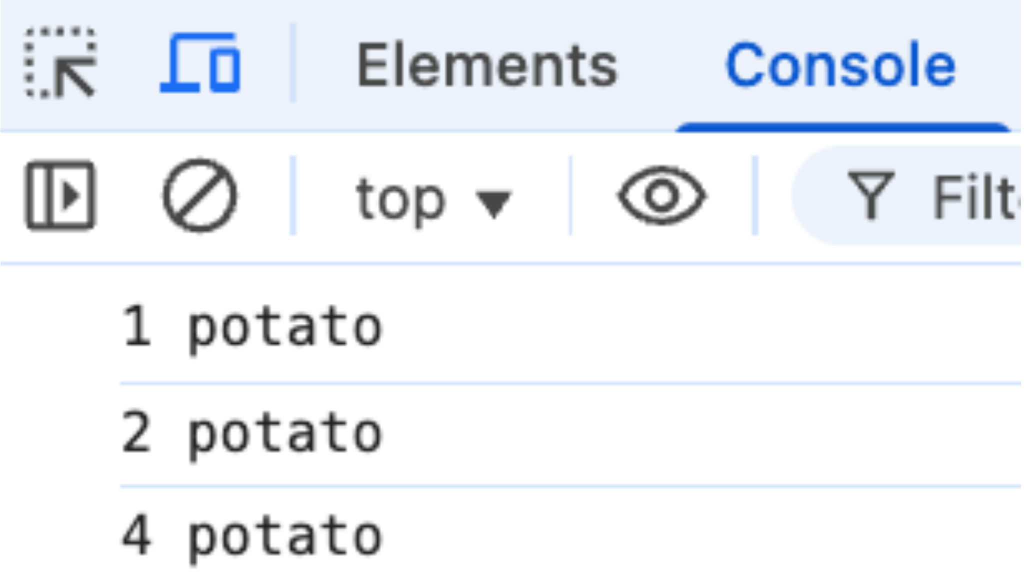


Variables

- Can be updated

- Can use all sorts of math functions

```
let count = 1;
console.log(count + ' potato');
count = count + 1;
console.log(count + ' potato');
count = 2 * count;
console.log(count + ' potato');
```

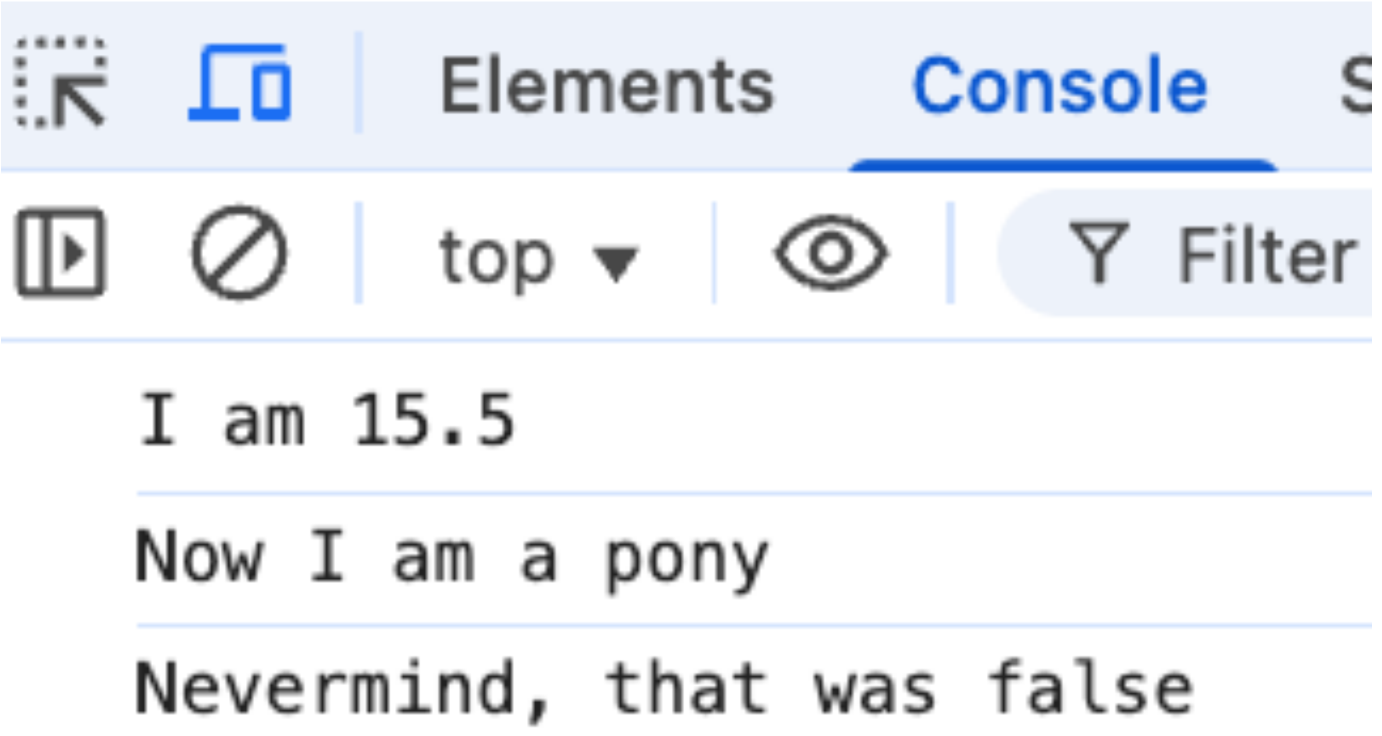


Variables

- Variables have *types*
    - Numbers (whole numbers and decimals)
    - Strings (text)
    - Booleans (either true or false)
    - Many, many other types that we will ignore
- A variable can change between types in JavaScript

Variables

```javascript
let value = 15.5;
console.log('I am ' + value);
value = 'a pony';
console.log('Now I am ' + value);
value = false;
console.log('Nevermind, that was ' + value);
```



Variables

1 fish	solution2.js:3
2 fish	solution2.js:7
red fish	solution2.js:11
blue fish	solution2.js:15

Conditionals

Conditionals

- Often, you want some code to run only *if* a particular condition is met
- Brackets `{ }` indicate blocks of code to run
- Can optionally designate other code to run *else* condition is not met

```
let isEighteen = true;
if(isEighteen) {
    console.log("I can vote!");
} else {
    console.log("Need to wait to vote");
}
```

Conditionals

- Can chain `if`s with `else if`
- Brackets `{ }` can run multiple lines of code

```
let myAge = 23;
if(myAge < 18) {
    console.log("Need to wait to vote");
} else if (myAge < 21) { //We know I am at least 18
    console.log("I can vote!");
    console.log("But I need to wait to drink");
} else { //We know I am at least 21
    console.log("I can vote and drink!");
}
```

Conditionals

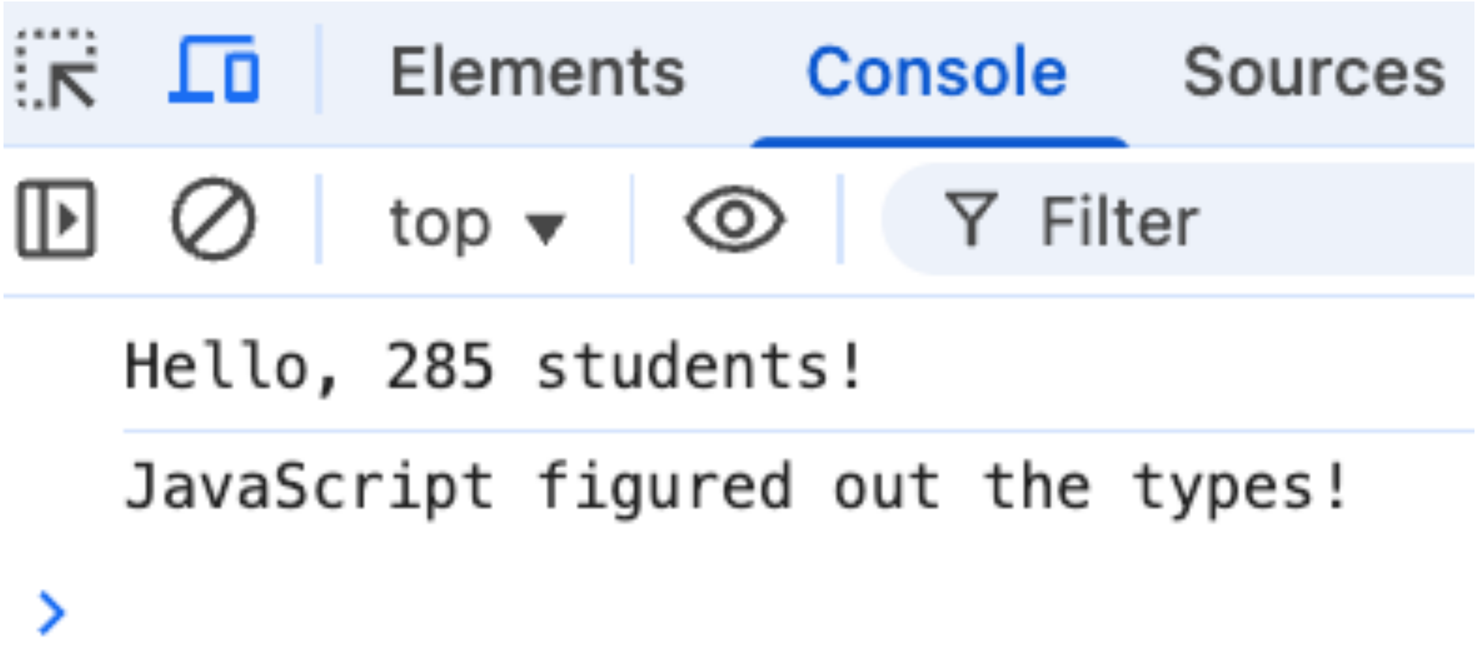
```
Jenny is 64        solution3.js:3
Not eligible       solution3.js:8
Jenny is 65        solution3.js:12
Eligible           solution3.js:15
```

Conditionals

- Equals can be checked with ==

- JavaScript will try to convert between types

```
let courseNumber = 285;
if(courseNumber == 285) {
    console.log('Hello, 285 students!');
}
if(courseNumber == '285') {
    console.log('JavaScript figured out the types!');
}
```



Conditionals

- “Not” can be specified with an !

- Either checking “not equal” or inverting a boolean value

```
let courseNumber = 285;
if(courseNumber != 285) {
    console.log('Hello, other students!');
}
let isEighteen = false;
if(!isEighteen) {
    console.log('Need to wait to vote');
}
```

Conditionals

- Multiple conditionals can be combined with `&&` (**and**), `||` (**or**)

```
let age = 19;
let hasFakeId = true;
if(age >= 18 && age < 21) {
    console.log('I can vote, but not drink');
}

if(age >= 21 || hasFakeId) {
    console.log('I can drink... :-)');
}
```

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